

Changes to ThinkSystem tools

Details of the April 2019 changes to ThinkSystem tools

Lenovo

UEFI option changes



Click each number in turn to see more information about UEFI option changes.



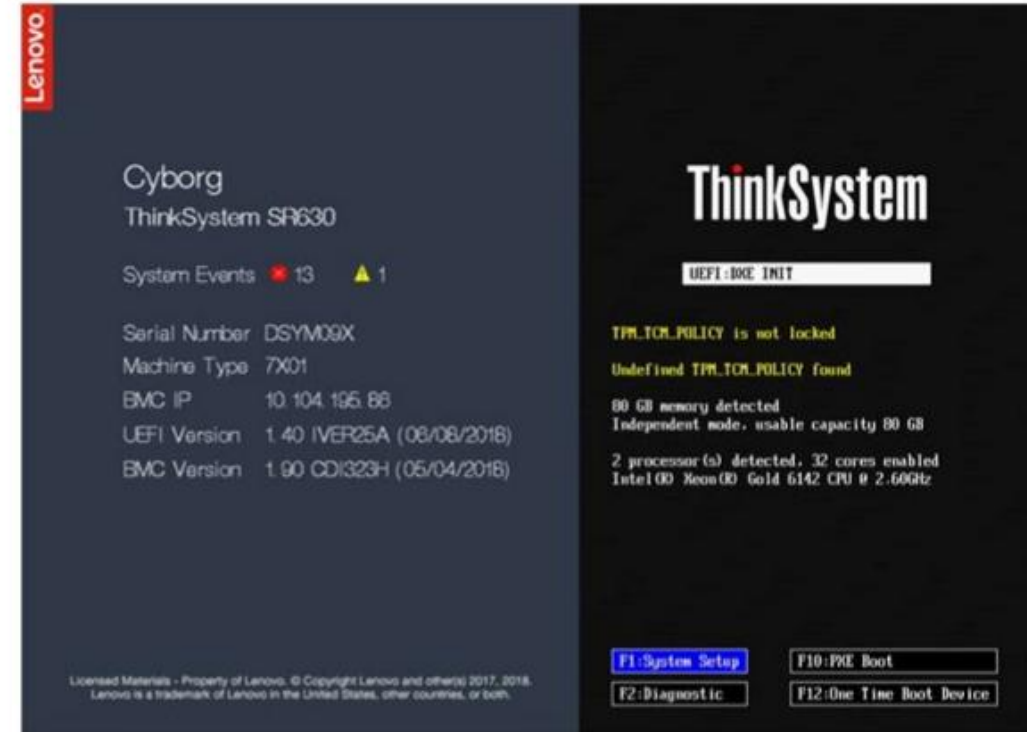
UEFI option changes

New options in boot menu

The latest version of LXPM supports two new options in the system boot menu:

- F2: Diagnostic
 - Press F2 to access the LXPM diagnostic menu
- F10: PXE boot

Click the image to enlarge.



UEFI option changes

No boot device and no bootable OS messages

If the system cannot find a boot device or bootable OS, the UEFI will display a **No boot device** message.

Click the buttons to see screenshots.

No boot device

No bootable OS



ThinkSystem SR650

System Events  1  1

Serial Number LSIT1810

Machine Type 7X05

BMC IP 10.104.197.116

UEFI Version 1.30 IVER350 (02/12/2019)

BMC Version 2.40 CDI333D (11/07/2018)



ThinkSystem

UEFI:POST END

Undefined TPM_TCM_POLICY found

32 GB memory detected
Independent mode, usable capacity 32 GB

2 processor(s) detected, 36 cores enabled
Intel(R) Xeon(R) Gold 6240C CPU @ 2.60GHz

Recalibrating System Power...

**No boot device is available, and the system will
reboot in about 5 seconds.**

F1: System Setup

F10: PXE Boot

F2: Diagnostic

F12: One Time Boot Device



No boot device is available, and the system will reboot in about 3 seconds.

UEFI option changes

VLAN configuration in Network Boot settings

In the past, the VLAN configuration options were listed on the Network page. In the latest UEFI, the VLAN configuration options are listed on the Network Boot Settings page.

Click [HERE](#) to see screenshots.





Network

ISCSI Settings

- ▶ Network Stack Settings
- ▶ Network Boot Settings

▲ Configure the iSCSI parameters.

Network Device List

- ▶ Slot 1: Emulex 10G NIC: Bus:Dev:Func 2F:0:0 - 00:90:FA:A2:0B:E2
- ▶ Slot 1: VLAN Configuration (MAC:0090FAA20BE2)
- ▶ Slot 1: Emulex 10G NIC: Bus:Dev:Func 2F:0:1 - 00:90:FA:A2:0B:EA
- ▶ Slot 1: VLAN Configuration (MAC:0090FAA20BEA)
- ▶ Slot 4: Broadcom NX-E PCIe 10Gb 2-Port Base-T Ethernet Adapter - 00:0A:F7:B0:EB:B0
- ▶ Slot 4: VLAN Configuration (MAC:000AF7B0EBB0)
- ▶ Slot 4: Broadcom NX-E PCIe 10Gb 2-Port Base-T Ethernet Adapter - 00:0A:F7:B0:EB:B1
- ▶ Slot 4: VLAN Configuration (MAC:000AF7B0EBB1)
- ▶ Slot 4: VLAN Configuration (MAC:000AF7B0EBB3)
- ▶ Slot 4: VLAN Configuration (MAC:000AF7B0EBB5)
- ▶ Slot 4: VLAN Configuration (MAC:000AF7B0EBB7)
- ▶ Slot 3: QLogic QL41262 25Gb Eth
- ▶ Slot 3: VLAN Configuration (MAC:00109B36C6EA)



Network Boot Settings

- ▶ MAC:00:10:9B:36:C6:EA Slot3 PFA AF:0:0
- ▶ MAC:00:10:9B:36:C6:F2 Slot3 PFA AF:0:1

Set the boot configuration parameters on MAC
00:10:9B:36:C6:EA

VLAN Configuration List:

- ▶ VLAN Configuration (MAC:00109B36C6EA)
- ▶ VLAN Configuration (MAC:00109B36C6F2)

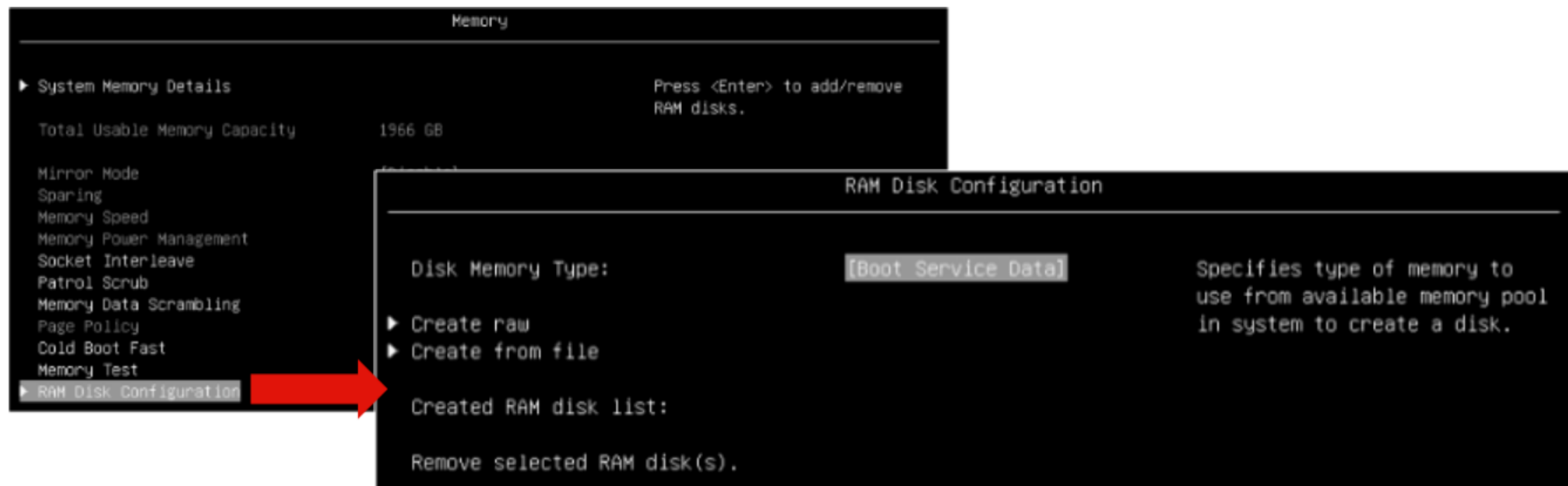
PCI Function Address: Bus
AF:Dev 0:Func 0

New

UEFI option changes

RAW disk function

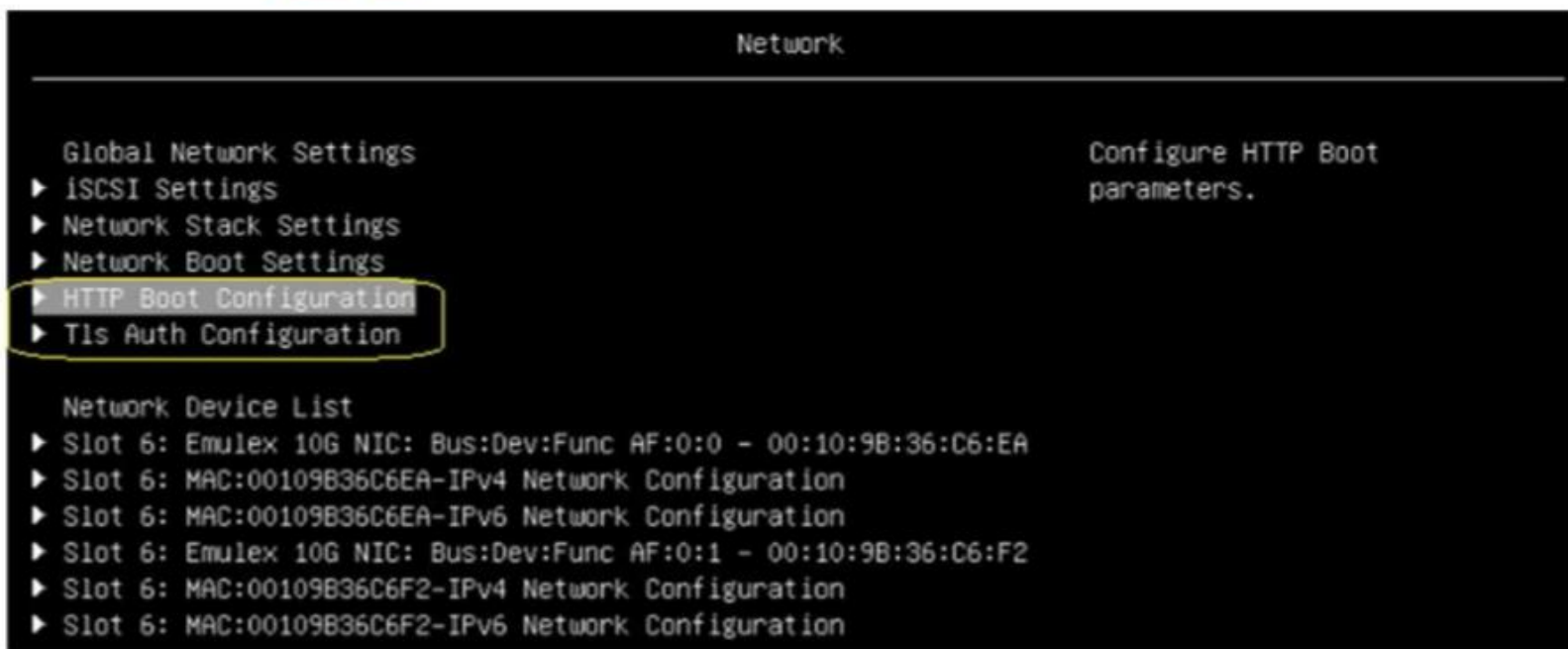
Users can now set up their RAW disk configuration through the UEFI. The RAW Disk Configuration option is listed on the Memory page.



UEFI option changes

HTTP Boot

The HTTP boot configuration and TLS authorization configuration options are listed on the Network page. Users can enable or disable HTTP boot on the Network → Network Stacking Configuration page. Click [HERE](#) to see a screenshot.





Network Stack Settings

Network Stack	[Enable]	Enable IPv4 HTTP Boot Support.
IPv4 PXE Support	[Enable]	If disabled IPv4 HTTP boot
IPv4 HTTP Support	[Disable]	option will not be created
IPv6 PXE Support	[Enable]	
IPv6 HTTP Support	[Disable]	
IPSEC Certificate	[Enable]	
PXE boot wait time	0	
Media detect count	1	



Network Stack Settings

Network Stack	[Enable]	Enable IPv4 HTTP Boot Support.
IPv4 PXE Support	[Enable]	If disabled IPv4 HTTP boot
IPv4 HTTP Support	[Disable]	option will not be created
IPv6 PXE Support	[Enable]	
IPv6 HTTP Support	[Disable]	
IPSEC Certificate	[Enable]	
PXE boot wait time	0	
Media detect count	1	

XCC changes

The latest version of XCC only supports inventory information for DCPMMs. Users cannot perform any other configuration actions on DCPMMs, such as changing operating mode or security states, through the XCC.

XClarity Controller | Chris' SR650 | System name: CG_Cyborg

DIMM: 3/24 Installed

Slot	Type	Capacity	Part Number
DIMM 5	DDR4	32 GB	M393A4K43BB2-CTD
DIMM 6	Optane DCPMM	512 GB	NMA1X8D512UQ5
DIMM 20	DDR4	32 GB	M393A4K43BB2-CTD

Description: DIMM_6		FRU Part Number	
FRU Serial Number	000022EE	Manufacturer	Intel
Type	Intel Optane DCPMM	Manufacture Date	3418 (wkly)
Max Speed	2666 MHz	Configured Memory Clock Speed	2666 MHz
Ecc Rmk	16	Module Supported Voltage	1.2V
Raw Capacity	502.5 GB	Memory Capacity	0 B
App Direct Capacity	0 B	Unconfigured Capacity	0 B
Inaccessible Capacity	502 GB	Reserved Capacity	61.3 MB
Firmware Version	01.00.00.5127		

Lenovo XClarity Essentials changes

Click the buttons to see the changes for each XClarity Essentials tool.

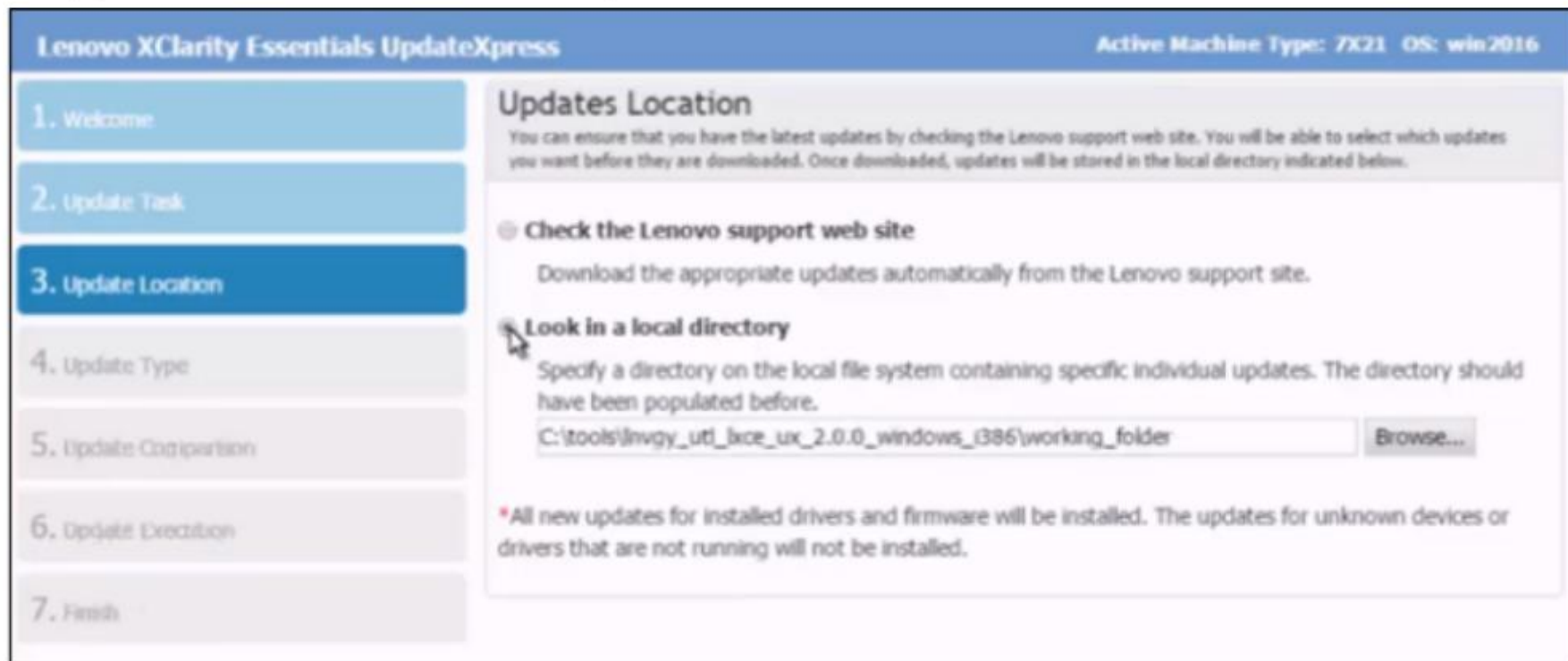
UpdateXpress

BoMC

OneCLI

UpdateXpress

- Supports firmware updates for ThinkSystem servers with second-generation Xeon Scalable processors [No Title]
- Supports DCPMM firmware updates



The screenshot displays the 'Lenovo XClarity Essentials UpdateXpress' application window. The title bar indicates 'Active Machine Type: 7X21 OS: win2016'. On the left, a vertical navigation pane lists seven steps: 1. Welcome, 2. Update Task, 3. Update Location (highlighted in blue), 4. Update Type, 5. Update Configuration, 6. Update Execution, and 7. Finish. The main content area is titled 'Updates Location' and includes a descriptive paragraph: 'You can ensure that you have the latest updates by checking the Lenovo support web site. You will be able to select which updates you want before they are downloaded. Once downloaded, updates will be stored in the local directory indicated below.' Below this, there are two radio button options. The first option, 'Check the Lenovo support web site', is selected and includes the instruction 'Download the appropriate updates automatically from the Lenovo support site.' The second option, 'Look in a local directory', is unselected and includes the instruction 'Specify a directory on the local file system containing specific individual updates. The directory should have been populated before.' Under the second option, there is a text input field containing the path 'C:\tools\invgy_uti_lxce_ux_2.0.0_windows_1386\working_folder' and a 'Browse...' button. At the bottom of the main area, a red asterisk note states: '*All new updates for installed drivers and firmware will be installed. The updates for unknown devices or drivers that are not running will not be installed.'

Lenovo XClarity Essentials changes

Click the buttons to see the changes for each XClarity Essentials tool.

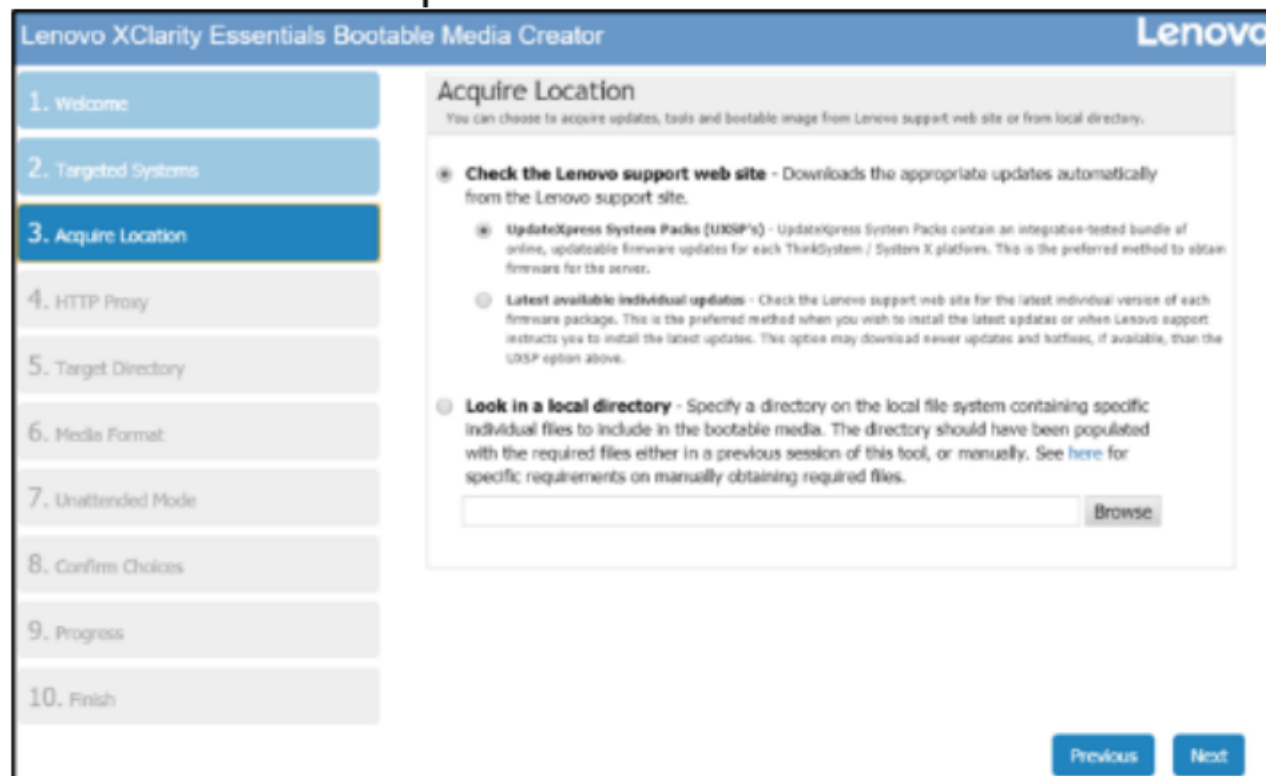
UpdateXpress

BoMC

OneCLI

BoMC

- Supports ThinkSystem servers with second-generation Xeon Scalable processors
- Supports DCPMM firmware updates



Lenovo XClarity Essentials changes

Click the buttons to see the changes for each XClarity Essentials tool.

UpdateXpress

BoMC

OneCLI

OneCLI

- For the latest OneCLI user guide, go to the OneCLI Lenovo Support Web site: <https://datacentersupport.lenovo.com/tw/en/solutions/invo-tcli>.
- Supports remote batch firmware updates (`multiflash` command). For more information, refer to the *OneCLI User Guide – Updating firmware for multiple BMC servers remotely* section.
- Supports software RAID configuration with Intel RSTe Software on Linux. The software RAID configuration with Intel RSTe Software supports the following commands: `show`, `add`, and `clear`. For more information, refer to the *OneCLI User Guide – raid command* section.
- Supports DCPMM firmware updates and inventory functions. For more information, refer to ES51965 – Introducing the Intel DC Persistent memory module <https://lenovoedu.lenovo.com/course/view.php?idnumber=ES51965>.

Lenovo XClarity Provisioning Manager changes



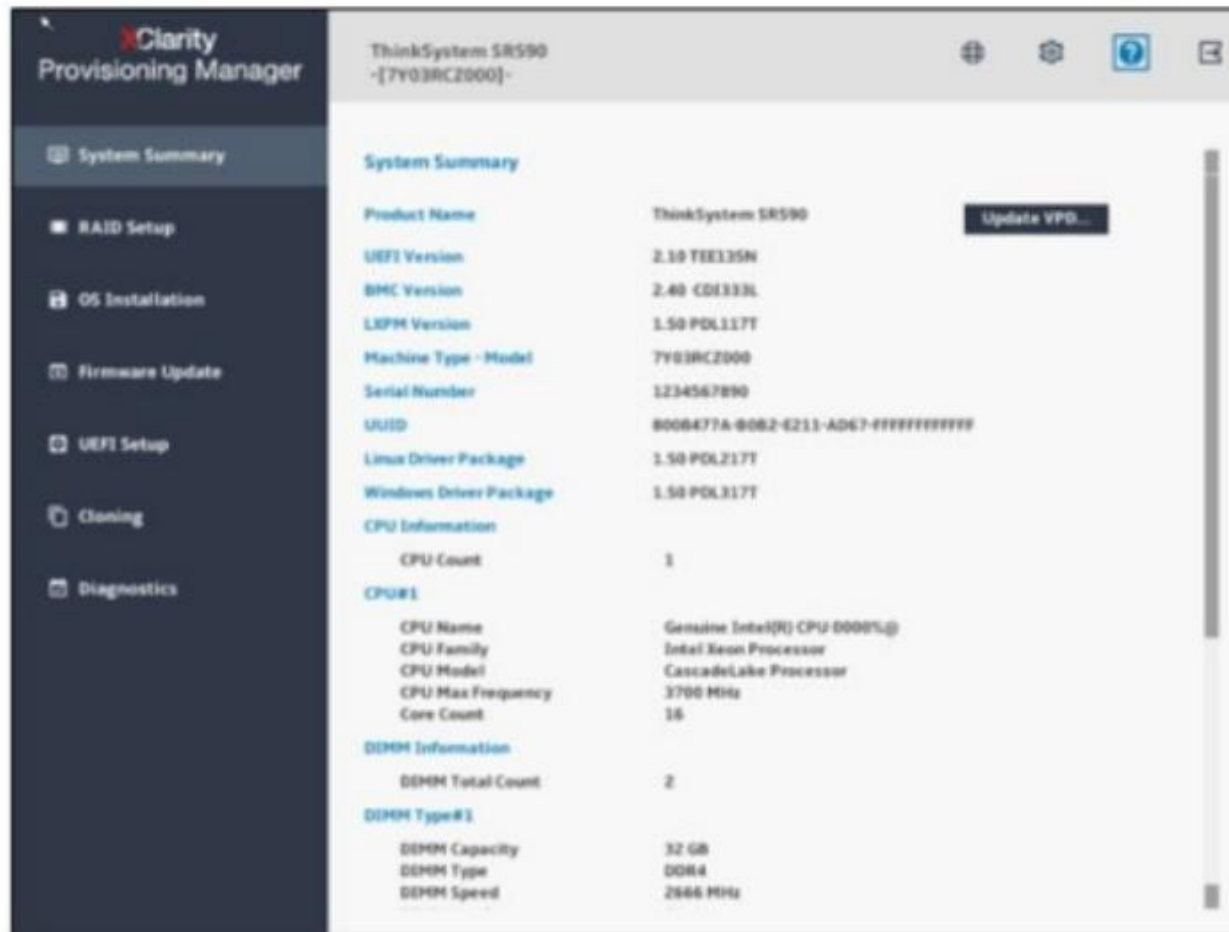
Click each number in turn to see more information about the Lenovo XClarity Provisioning Manager (LXPM) changes.



Lenovo XClarity Provisioning Manager changes

System summary: LXPM can now display Xeon Scalable processor inventory information.

Click the image to enlarge.



Lenovo XClarity Provisioning Manager changes

Partition info: The LXPM system summary can also display system partition information.

The screenshot displays the XClarity Provisioning Manager interface for a ThinkSystem SR570. The left sidebar contains navigation options: System Summary, RAID Setup, OS Installation, Firmware Update, UEFI Setup, Cloning, and Diagnostics. The main content area shows system details for 'ThinkSystem SR570' with chassis ID '-[7Y02RCZ000]-'. It lists six storage models and their corresponding serial numbers. Below this, the 'Partition Info' section provides details for five partition types, including partition count, name, size, number, and format.

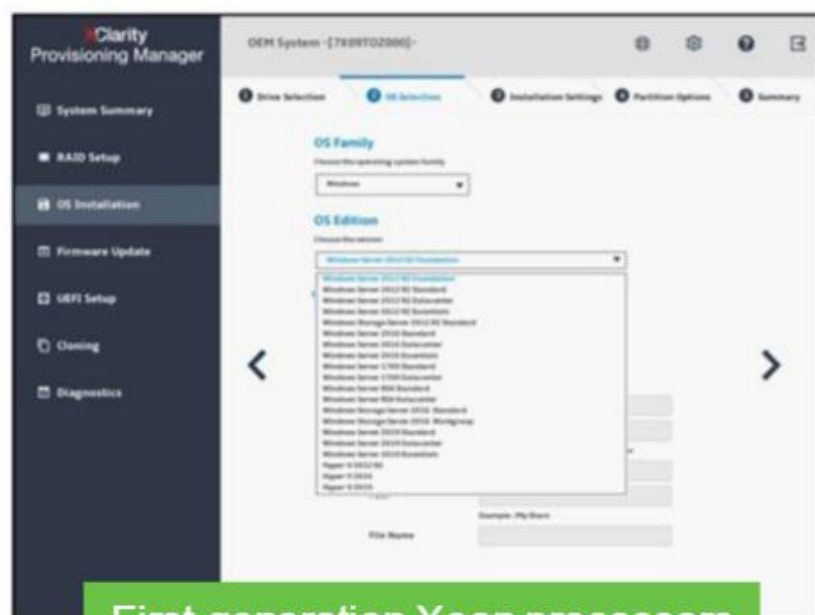
Model	SN
MODEL#:ST4000NM0025	SN#:ZC100FAY0000C7099UY5
MODEL#:SU84ENC5DR24C	SN#:BP51014687G005F
MODEL#:ST4000NM0025	SN#:ZC100IE20000C7099T9J
MODEL#:HUS156060VLS600	SN#:JZWX2Y6M
MODEL#:ST4000NM0025	SN#:ZC100FFA0000C7099UWJ
MODEL#:ST4000NM0025	SN#:ZC10CED40000R7105X42
MODEL#:ST4000NM0025	SN#:ZC10DFGE0000C7099V1K

Partition Type	Partition Name	Partition Size	Partition Number	Partition#	Partition#
Partition Type#1	Volume 0	3726 GB	2	GPT	FAT
Partition Type#2	Volume 1	558 GB	0		
Partition Type#3	Volume 2	558 GB	1	GPT	
Partition Type#4	Volume 3	558 GB	0		
Partition Type#5					

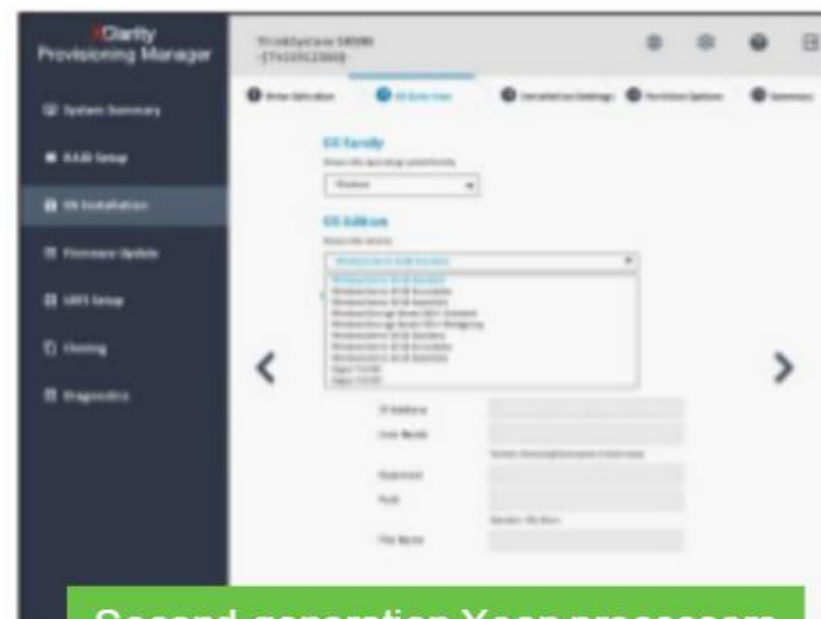


Lenovo XClarity Provisioning Manager changes

OS installation: The latest version of LXPM supports new OSs for ThinkSystem servers with first- and second-generation Xeon processors. Click the images to enlarge.



First-generation Xeon processors



Second-generation Xeon processors



Lenovo XClarity Provisioning Manager changes

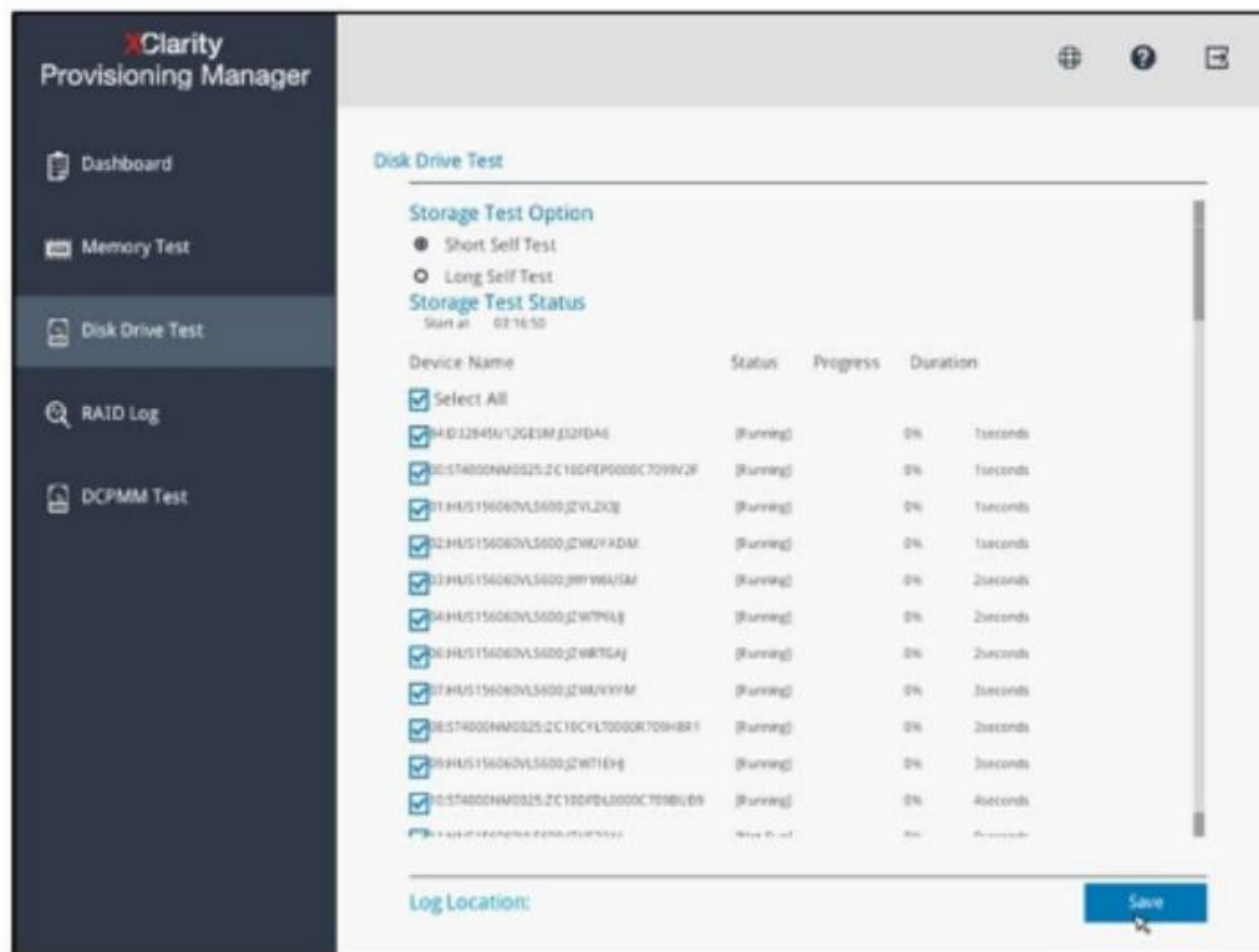
Disk drive test

LXPM diagnostics now supports the SMART drive test. Users can run the SMART test through the LXPM and then click Save to export the log file.

Click the buttons for disk drive test log result samples.

SAS/SATA drive log

NVMe drive log





Scroll down for more information.

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

===== 01:ST9500620NS:9XF2QTSL =====Port=1

=== START OF INFORMATION SECTION ===

Device Model: ST9500620NS

Serial Number: 9XF2QTSL

LU WWN Device Id: 5 000C50 065FCC31A

Firmware Version: SN03

User Capacity: 500 GB

Sector Size: 512 bytes logical/physical

Rotation Rate: 7200 rpm

Form Factor: 2.5 inches

ATA Version is: ATA8-ACS T13/1699-D revision 4

SATA Version is: SATA 3.0, 6.0 Gb/s (current: 6.0 Gb/s)

General SMART Values:

Offline data collection status: (0x82) Offline data collection activity
was completed without error.

Auto Offline Data Collection: Enabled.

Self-test execution status: (0) The previous self-test routine completed
without error or no self-test has ever
been run.

Total time to complete Offline
data collection: (642) seconds.

Offline data collection
capabilities: (0x7B) SMART execute Offline immediate.
Auto Offline data collection on/off support.
Suspend Offline collection upon new command.
Offline surface scan supported.
Self-test supported.



Scroll down for more information.

```
Self-test execution status:      (  0) The previous self-test routine completed
                                  without error or no self-test has ever
                                  been run.
Total time to complete Offline
data collection:                ( 642) seconds.
Offline data collection
capabilities:                    (0x7B) SMART execute Offline immediate.
                                  Auto Offline data collection on/off support.
                                  Suspend Offline collection upon new command.
                                  Offline surface scan supported.
                                  Self-test supported.
                                  Conveyance Self-test supported.
                                  Selective Self-test supported.
SMART capabilities:             (0x0003) Saves SMART data before entering power-saving
                                  Supports SMART auto save timer.
Error logging capability:       (0x01) Error logging supported.
                                  General Purpose Logging supported.
Short self-test routine
recommended polling time:      (  1) minutes.
Extended self-test routine
recommended polling time:      (102) minutes.
Conveyance self-test routine
recommended polling time:      (  2) minutes.
```

[No Title]



Scroll down for more information.

```
文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)
===== 00:SSDPE2KE016T7L:BTLE821008EZ1P6CGN =====Port=0
NVMe ctrl=913940A8
=== START OF INFORMATION SECTION ===
Model Number:          SSDPE2KE016T7L
Serial Number:         BTLE821008EZ1P6CGN
Firmware Version:      QDV1LV45
PCI Vendor ID:         0x8086
PCI Vendor Subsystem ID: 0x1D49
IEEE OUI Identifier:    0x5CD2E4
Controller ID:         0
=== START OF SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED

Critical Warning:      0x00
Temperature:          47 Celsius
Available Spare:       100%
Available Spare Threshold: 10%
Percentage Used:       0%
Data Units Read:       1 [512KB]
Data Units Written:    0
Host Read Commands:    278
Host Write Commands:   0
Controller Busy Time:  0
Power Cycles:          18
Power On Hours:        8
Unsafe Shutdowns:      16
Media and Data Integrity Errors: 0
Error Information Log Entries: 0
Warning Comp. Temperature Time: 0
Critical Comp. Temperature Time: 0
```




Scroll down for more information.

SMART overall-health self-assessment test result: PASSED

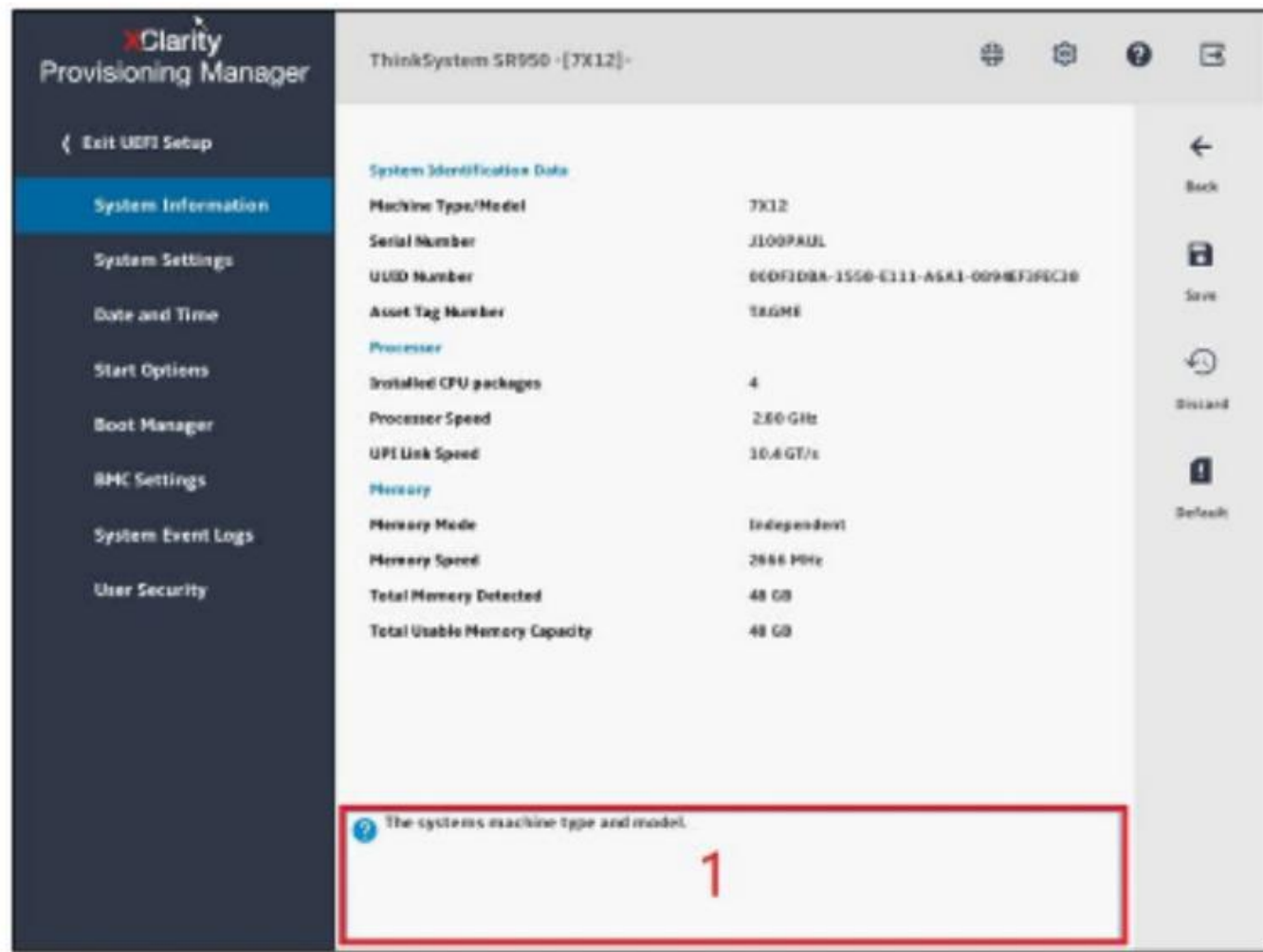
Critical Warning:	0x00
Temperature:	47 Celsius
Available Spare:	100%
Available Spare Threshold:	10%
Percentage Used:	0%
Data Units Read:	1 [512KB]
Data Units Written:	0
Host Read Commands:	278
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Controller Busy Time:	0
Power Cycles:	18
Power On Hours:	8
Unsafe Shutdowns:	16
Media and Data Integrity Errors:	0
Error Information Log Entries:	0
Warning Comp. Temperature Time:	0
Critical Comp. Temperature Time:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Error Information Log Entries:	0
Thermal Temp. 1 Transition Count:	0

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Lenovo XClarity Provisioning Manager changes

UEFI setup

In the UEFI setup menu, LXPM can now display a help message for each option. These help messages are the same as the messages in the UEFI setup text mode.



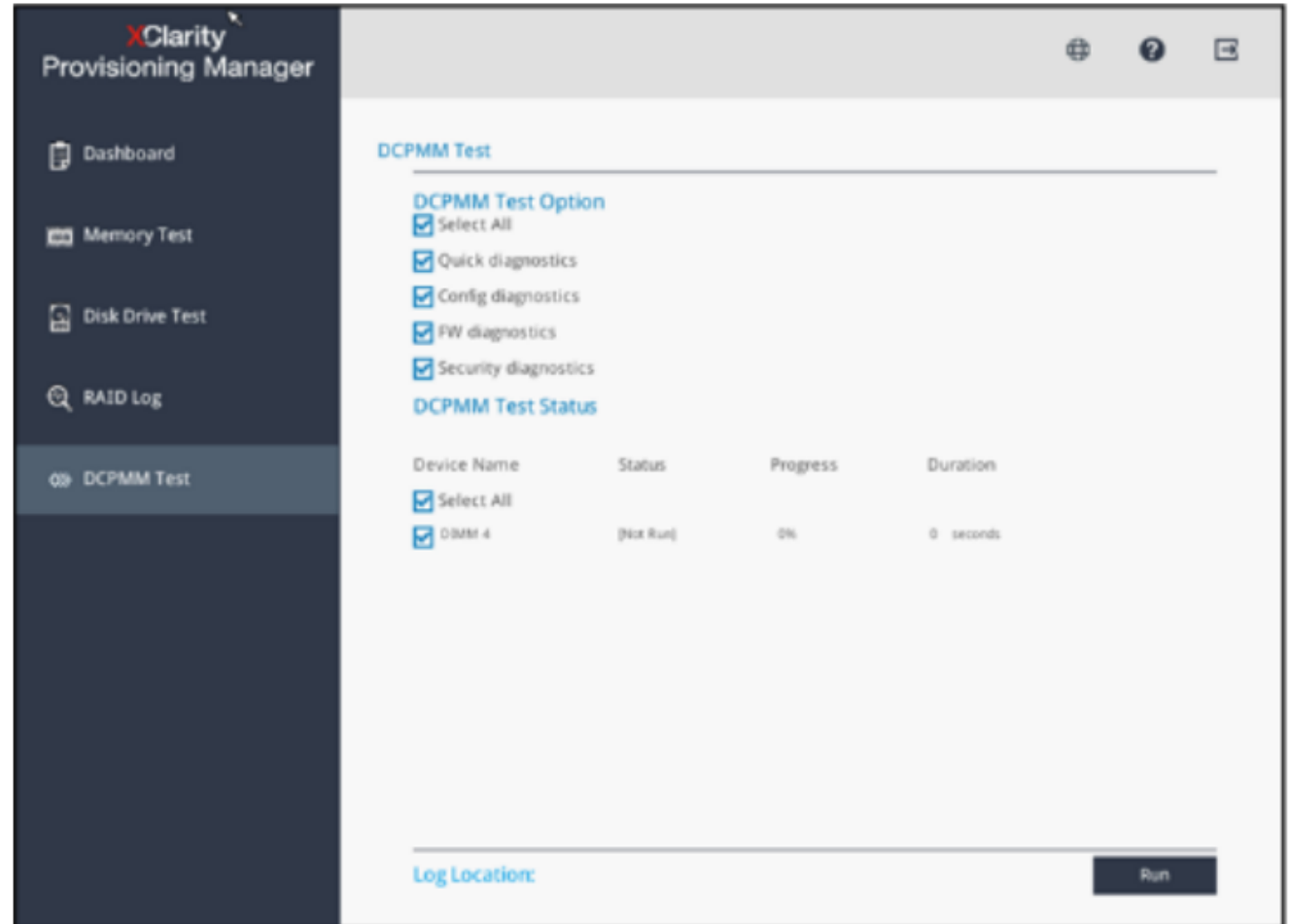
Lenovo XClarity Provisioning Manager changes

DCPMM test

If the DCPMMs on the system are set to App Direct mode or Mixed Memory mode, users can run a DCPMM test for DCPMM hardware diagnostics.

If the DCPMMs are provisioned for Memory mode, use a Memory Test for both DIMM and DCPMM hardware diagnostics. For more information, refer to ES51965 – Introducing the Intel DC Persistent memory module

<https://lenovoedu.lenovo.com/course/view.php?idnumber=ES51965>.



Lenovo XClarity Administrator changes

The latest version of XClarity Administrator supports DCPMM inventory monitoring and security operation settings. For more information, refer to ES51965 – Introducing the Intel DC Persistent memory module <https://lenovoedu.lenovo.com/course/view.php?idnumber=ES51965>.

The screenshot displays the 'Memory: (4) Installed' section of the XClarity Administrator. A table lists four memory modules. The first module, Bay 5, is an Intel Optane DC Persistent Memory module (256 GB, 2666 MT/s). A dashed blue box highlights this entry, and a pop-up window titled 'Intel Optane Summary' is open over it. The pop-up provides detailed information about the Optane module's capacity and firmware. To the right, a 'Manufacturer' list shows Intel, Micron Technology, and Samsung.

Bay Number	Size	Speed	Type
5	256 GB	2666 MT/s	Intel Optane DC...
6	16 GB	2666 MT/s	DDR4
7	16 GB	2666 MT/s	DDR4
8	16 GB	2666 MT/s	DDR4

Intel Optane Summary

Memory Direct Capacity	252 GB
Memory Inaccessible Capacity	448 MB
Memory Capacity	0 MB
Firmware Name	AEP DIMM firmware
Firmware Version	01.02.00.5318
Firmware Status	Active

Manufacturer
Intel
Micron Technology
Samsung
Samsung

Summary

This course enabled you to:

- Describe the features of Intel second-generation Xeon Scalable processors
- Describe the updated ThinkSystem server specifications
- Describe the updated ThinkSystem tool features